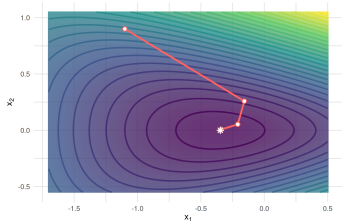


Optimization in Machine Learning

Second order methods

Newton-Raphson vs Gradient Descent



Learning goals

- Comparison of Newton-Raphson and Gradient Descent
- Pure Newton vs relaxed Newton with step size

NEWTON-RAPHSON AND GD (RECAP)

- Gradient Descent: **first order method**
⇒ *Gradient* information, i.e., first derivatives
- Newton-Raphson: **second order method**
⇒ *Hessian* information, i.e., second derivatives

Gradient Descent:

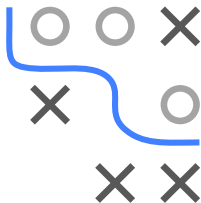
$$\theta^{[t+1]} = \theta^{[t]} - \alpha \nabla f(\theta^{[t]})$$

Pure Newton-Raphson:

$$\theta^{[t+1]} = \theta^{[t]} - (\nabla^2 f(\theta^{[t]}))^{-1} \nabla f(\theta^{[t]})$$

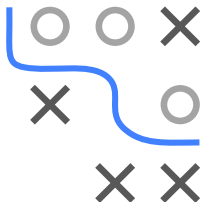
Relaxed/Damped Newton-Raphson:

$$\theta^{[t+1]} = \theta^{[t]} - \alpha (\nabla^2 f(\theta^{[t]}))^{-1} \nabla f(\theta^{[t]})$$



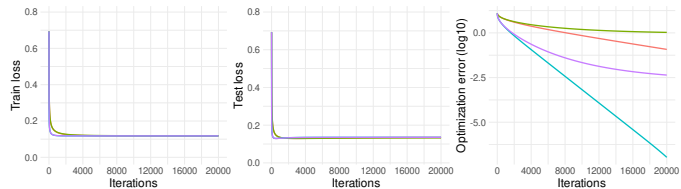
COMPARISON SIMULATION SET-UP

- Comparison of Newton-Raphson, relaxed NR and GD+momentum:
- **Logistic regression** (log loss) simulation with $n = 500$ samples and $p = 11$ features, where $\theta^* = (-5, -4, \dots, 0, \dots, 4, 5)^T$, and $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \Sigma)$ for $\Sigma = \mathbf{I}$ (i.i.d.) or $\Sigma_{i,j} = 0.99^{|i-j|}$ (corr. features)
- To simulate response, we set $y^{(i)} \sim \mathcal{B}(\pi^{(i)})$, $\pi^{(i)} = \frac{1}{1 + e^{-(\mathbf{x}^{(i)})^T \theta^*}}$
- Indep. features result in a condition number of ≈ 2.9 while corr. features yield (moderately) bad condition number ≈ 600
- ERM has unique global minimizer (convexity) but no closed-form solution. We can approximate $\hat{\theta}$ using `glm` solution
- We also track the optimization error $\|\theta - \hat{\theta}\|_2$
- For relaxed NR we use $\alpha = 0.7$ and for GD we set $\alpha = 1$, momentum to 0.8 and use no step size control

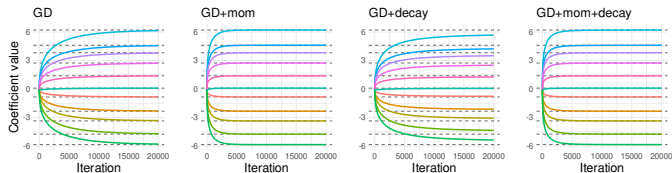


LOGISTIC REGRESSION (GD VARIANTS RECAP)

- Recall comparison of GD variants on log. reg. in last chapter:

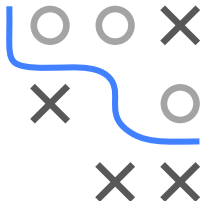


GD — GD+decay — GD+mom — GD+mom+decay Method — GD — GD+decay — GD+mom



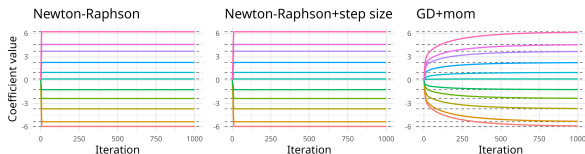
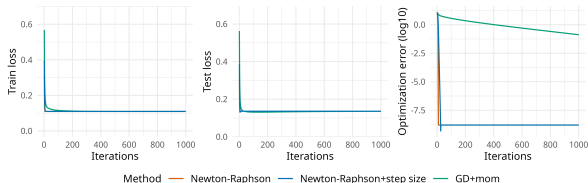
Dotted lines indicate global minimizers.

- GD+momentum** was fastest $\Rightarrow$ now compare w/ Newton-Raphson
- NB:** GD+momentum only converges after several thousand steps



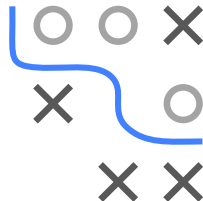
LOGISTIC REGRESSION (GD VS. NR)

- Let's run GD vs. NR for 1000 **steps** (independent features):



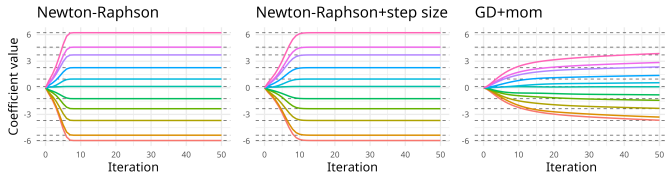
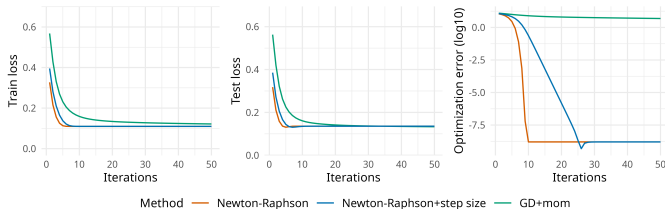
Dotted lines indicate global minimizers.

- NR** and **relaxed NR** $\Rightarrow$ almost instantaneous convergence (see optimization error)
- Using $\alpha < 1$ slightly slows down **relaxed NR**
- GD+mom** several orders of magnitude slower than NR



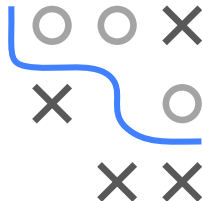
LOGISTIC REGRESSION (GD VS. NR)

- Let's run the same configuration only for 50 **steps** to see clearer picture:



Dotted lines indicate global minimizers.

- NR** takes ≈ 10 steps to reach same optimization error as **GD+mom** after 20,000 steps!
- Relaxed NR** with $\alpha < 1$ shows no advantage here



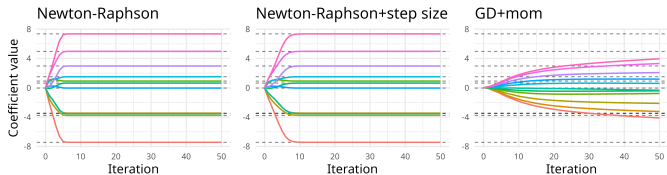
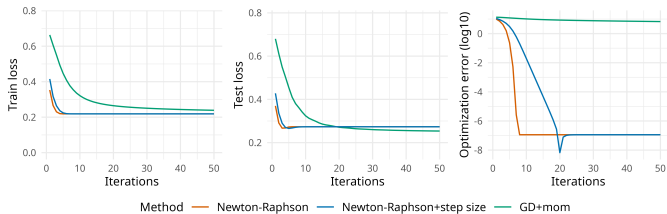
A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are empty. The cells (0,1), (1,1), and (2,2) contain a grey 'X'. The cells (1,0) and (2,1) contain a grey circle.

-
- The figure consists of two side-by-side line plots. The left plot is titled 'Training loss' and the right plot is titled 'Parameter optimization error'. Both plots share the same x-axis, 'Seconds', ranging from 0.00 to 0.06. The legend at the bottom indicates three methods: Newton-Raphson (orange line), Newton-Raphson+step size (blue line), and GD+mom (green line).
- Training loss plot:** The y-axis ranges from 0.1 to 0.5. Newton-Raphson+step size (blue) starts at ~0.4 and drops to ~0.1 by 0.01s. Newton-Raphson (orange) starts at ~0.32 and drops to ~0.1 by 0.025s. GD+mom (green) starts at ~0.55 and drops to ~0.1 by 0.04s.
- Parameter optimization error plot:** The y-axis ranges from -7.5 to 0.0. Newton-Raphson+step size (blue) starts at ~0.0 and drops to ~-8.5 by 0.025s. Newton-Raphson (orange) starts at ~0.0 and drops to ~-8.5 by 0.025s. GD+mom (green) starts at ~0.0 and drops to ~-8.5 by 0.04s.
- | Seconds | Newton-Raphson (Loss) | Newton-Raphson+step size (Loss) | GD+mom (Loss) | Newton-Raphson (Error) | Newton-Raphson+step size (Error) | GD+mom (Error) |
|---------|-----------------------|---------------------------------|---------------|------------------------|----------------------------------|----------------|
| 0.00 | - | 0.40 | 0.55 | 0.0 | 0.0 | 0.0 |
| 0.01 | 0.32 | 0.12 | 0.25 | 0.0 | -1.0 | 0.0 |
| 0.02 | 0.22 | 0.11 | 0.15 | 0.0 | -4.0 | 0.0 |
| 0.025 | 0.11 | 0.11 | 0.13 | -8.5 | -8.5 | -8.5 |
| 0.04 | 0.11 | 0.11 | 0.12 | -8.5 | -8.5 | -8.5 |

- **NR** steps are indeed slower than **GD** steps ($\approx 3\times$ here)
- But each NR step is so much better than GD ($\approx 2000\times$) that per-iteration runtime advantage of GD becomes **irrelevant**

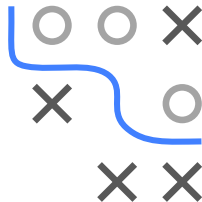
LOGISTIC REGRESSION (CORR.)

- In case of correlated features the results are very similar:



Dotted lines indicate global minimizers.

- **NR's** performance is unaffected by feature correlation
- **GD** iterates become “warped” compared to before



A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are empty. The cells (0,1), (1,1), and (2,2) contain a grey 'X'. The cells (1,0) and (2,1) contain a grey circle.

-
- The figure consists of two side-by-side line plots. The left plot, titled 'Training loss', shows the training loss on the y-axis (ranging from 0.2 to 0.6) against time in seconds on the x-axis (ranging from 0.00 to 0.04). The right plot, titled 'Parameter optimization error', shows the parameter optimization error (log10) on the y-axis (ranging from -8 to 0) against time in seconds on the x-axis (ranging from 0.00 to 0.04). Both plots compare three methods: Newton-Raphson (orange line), Newton-Raphson+step size (blue line), and GD+mom (green line). In the training loss plot, GD+mom starts at ~0.65 and decreases to ~0.24. Newton-Raphson+step size starts at ~0.42 and decreases to ~0.22. Newton-Raphson starts at ~0.35 and decreases to ~0.22. In the parameter optimization error plot, GD+mom remains near 0. Newton-Raphson+step size starts at ~0.5 and decreases to ~-7.5, with a sharp dip to ~-8.0 around 0.015 seconds. Newton-Raphson starts at ~0.5 and decreases to ~-7.0, with a sharp dip to ~-7.5 around 0.005 seconds.
- | Seconds | Training loss (GD+mom) | Training loss (NR+step size) | Training loss (NR) | Param error (log10) (GD+mom) | Param error (log10) (NR+step size) | Param error (log10) (NR) |
|---------|------------------------|------------------------------|--------------------|------------------------------|------------------------------------|--------------------------|
| 0.00 | 0.65 | 0.42 | 0.35 | 0.5 | 0.5 | 0.5 |
| 0.005 | 0.50 | 0.22 | 0.22 | 0.4 | -7.5 | -7.0 |
| 0.01 | 0.30 | 0.22 | 0.22 | 0.3 | -7.5 | -7.0 |
| 0.015 | 0.26 | 0.22 | 0.22 | -8.0 | -7.5 | -7.0 |
| 0.02 | 0.25 | 0.22 | 0.22 | -7.5 | -7.5 | -7.0 |
| 0.04 | 0.24 | 0.22 | 0.22 | -7.5 | -7.5 | -7.0 |

- **NR** steps are indeed slower than **GD** steps ($\approx 4\times$ here)
- Overall **NR** is strongly superior to **GD** wrt optim error and speed